

Basic Principles of Medicine 1

Foundations to Medicine

FTM 23: Clinical Enzymology

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Objectives: Clinical enzymology

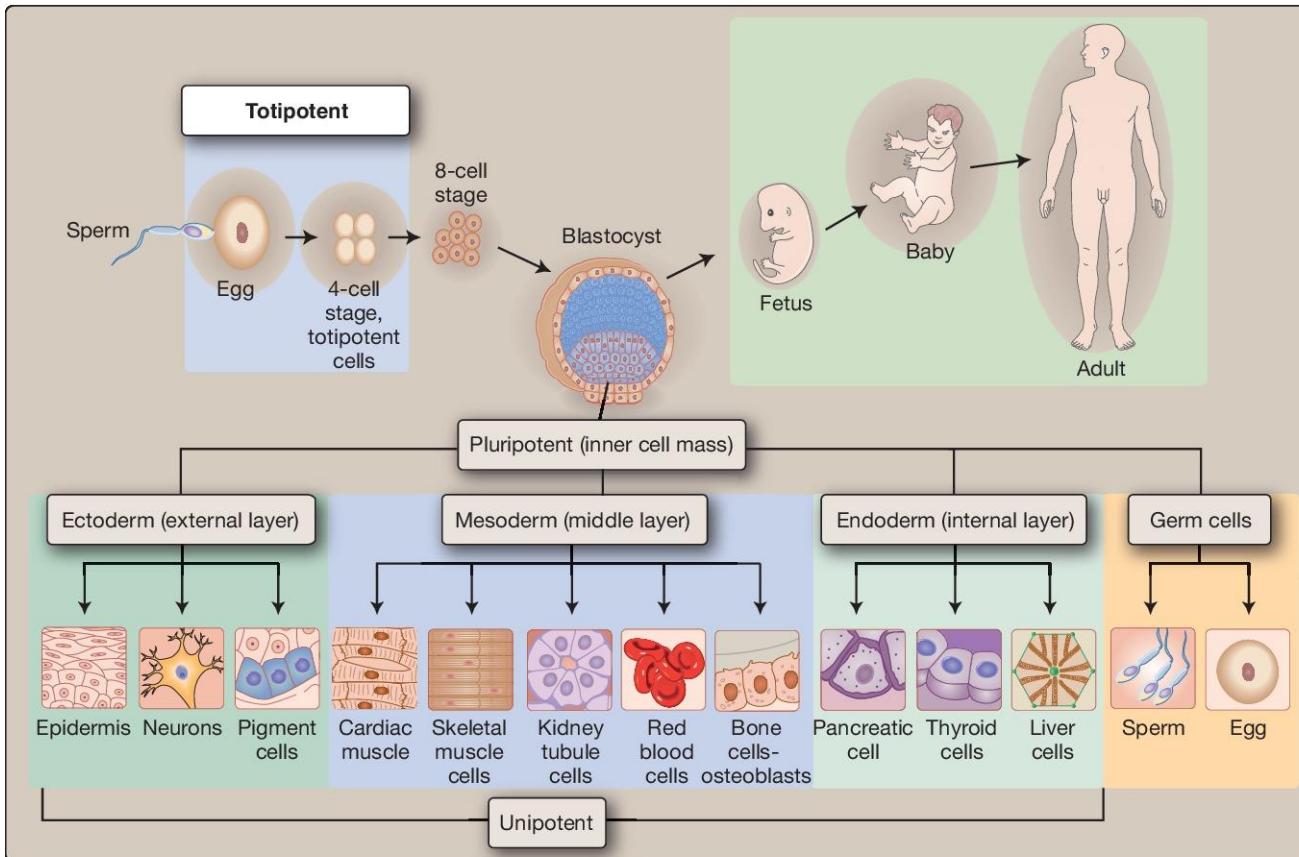
SOM.MK.I.BPM1.1.FTM.3.BCHM.0136	Review the significance of estimation of LDH.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0137	Differentiate isozymes of creatine kinase based on tissue location and subunit composition.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0138	Discuss utility of serum injury markers following myocardial infarction: Myoglobin, CK-MB/total CK ratio, serum cardiac troponins I and T
SOM.MK.I.BPM1.1.FTM.3.BCHM.0139	Discuss the utility of serum injury markers following MI: Time frame of serum injury markers (myoglobin, cardiac troponins, CK-MB) with the help of a graph.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0140	Describe the application of ALT, AST, ALP and GGT as markers of hepatocellular disease, biliary disease and alcohol liver disease.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0141	Discuss significance of serum amylase and serum lipase related to pancreatic disease.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0142	Indicate specific serum injury markers for bone disease (ALP), prostate cancer (PSA) and liver cancer (alpha-fetoprotein).

Recommended Reading

- Lippincott's Biochemistry 9th Edition_ 
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>; Chapter 5: Enzymes
- Enzymes | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
- Cardiac Markers in the Early Diagnosis and Management of Patients with Acute Coronary Syndrome. A.A. Al-Hadi and K.A. Fox SQU Med.J. Dec 2009 9(3) 231-236:
[http:// www.ncbi.nlm.nih.gov/pmc/articles/PMC3074795](http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3074795)

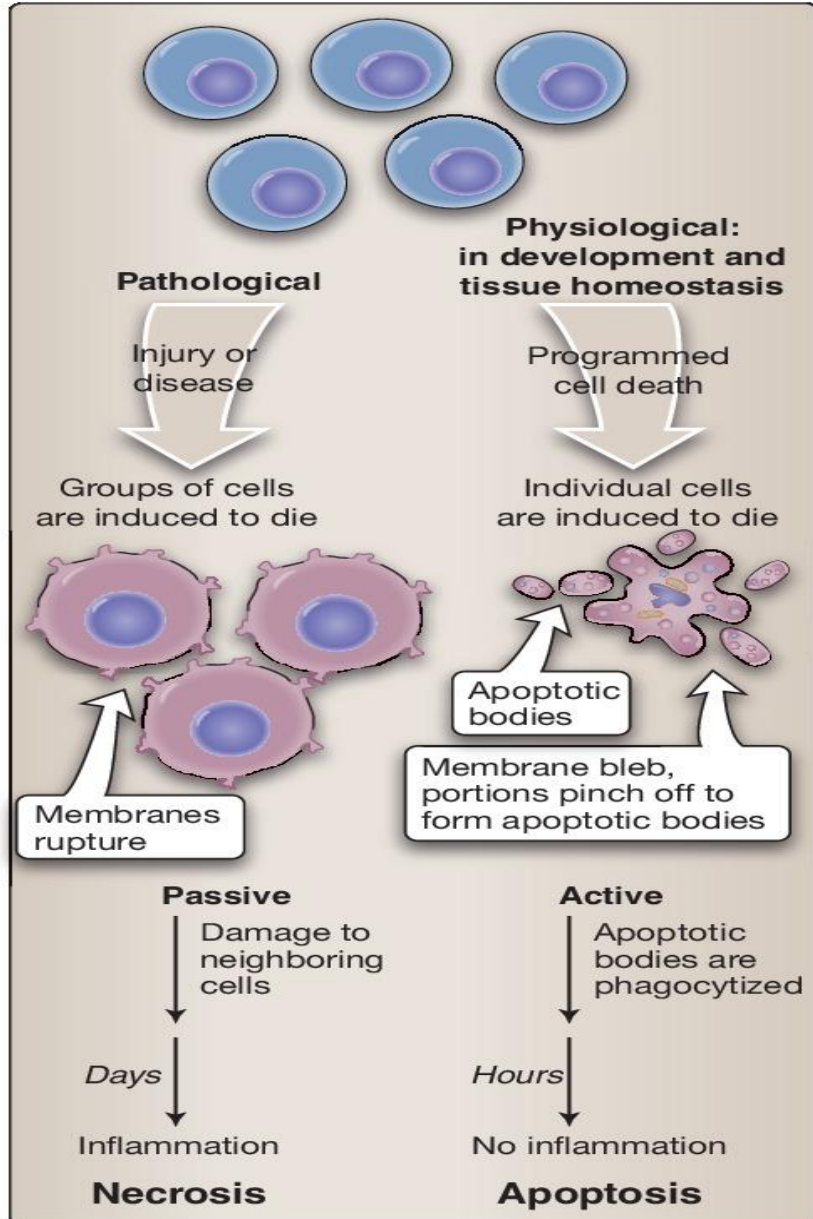
Cell differentiation

- All cells contain the same inherited nuclear DNA.
- Differentiation: cells specialize in their determined functions
- Structural and chemical differences
- **Specific enzymes are characteristic of specific cells**

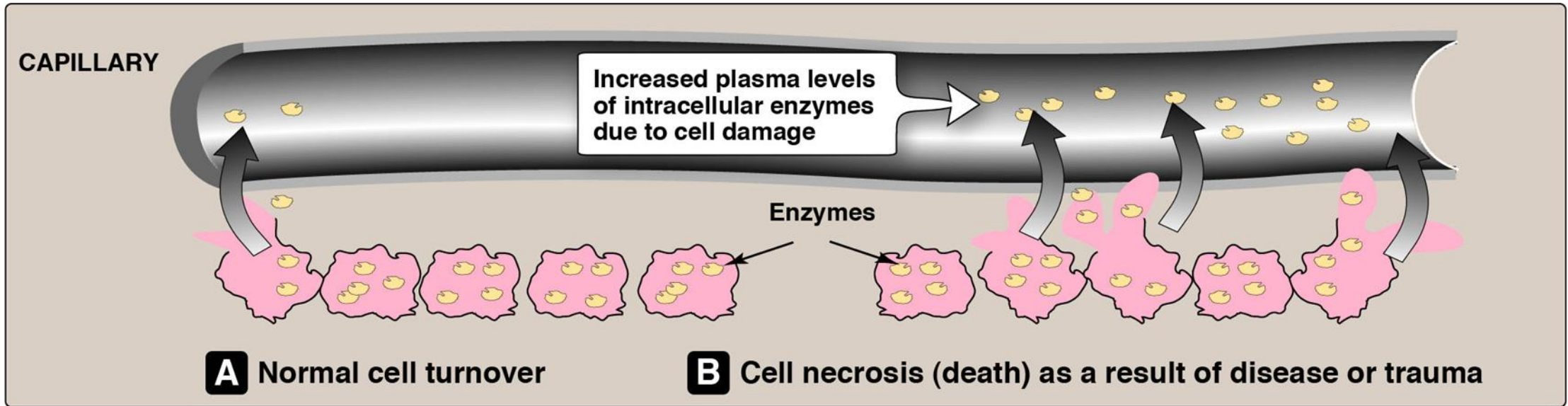


Cell death

- Most enzymes function intracellularly, but some can be found in serum
- Some enzymes are secreted into blood and have different functions.
 - Enzymes released during normal cell turnover (Apoptosis)
 - Enzymes leak into blood due to inflammation or cell death or pathology (**Inflammation**).



Intracellular enzyme release: Normal vs Pathology



- Increased plasma enzyme levels may indicate **increased cell damage**
- **Tissue damage** causing inflammation and/or cell lysis → Elevated enzyme levels
- Serum enzyme levels **correlates with extent or severity of tissue damage**

Clinically useful serum markers

Enzyme	Abbreviation	Main Tissue Source(s)	Useful to Assess
Lactate dehydrogenase	LDH	Red blood cells, liver, muscle— most cells	General marker of cell death; hemolysis
Troponins I or T	cTn (cTnI or cTnT)	Cardiac muscle	Cardiac muscle damage
Creatine kinase	CK	Muscle	Muscle damage or disease
Alanine aminotransferase	ALT	Liver	Liver damage or disease
Alkaline phosphatase	ALP	Bile duct, bone	Hepatobiliary disease (obstructive jaundice) and bone diseases
Amylase	Amylase	Pancreas	Pancreatic diseases
Aspartate aminotransferase	AST	Liver, muscle	Liver and muscle diseases
Gamma glutamyl transferase	GGT	Liver, bile duct	Hepatobiliary disease (obstructive jaundice)
Lipase	Lipase	Pancreas	Pancreatic diseases
Prostate specific antigen	PSA	Prostate	Prostate inflammation (malignancy, infection)
Alpha-fetoprotein	AFP	Liver	Liver malignancy

Hemolytic anemia: Lactate dehydrogenase (LDH)

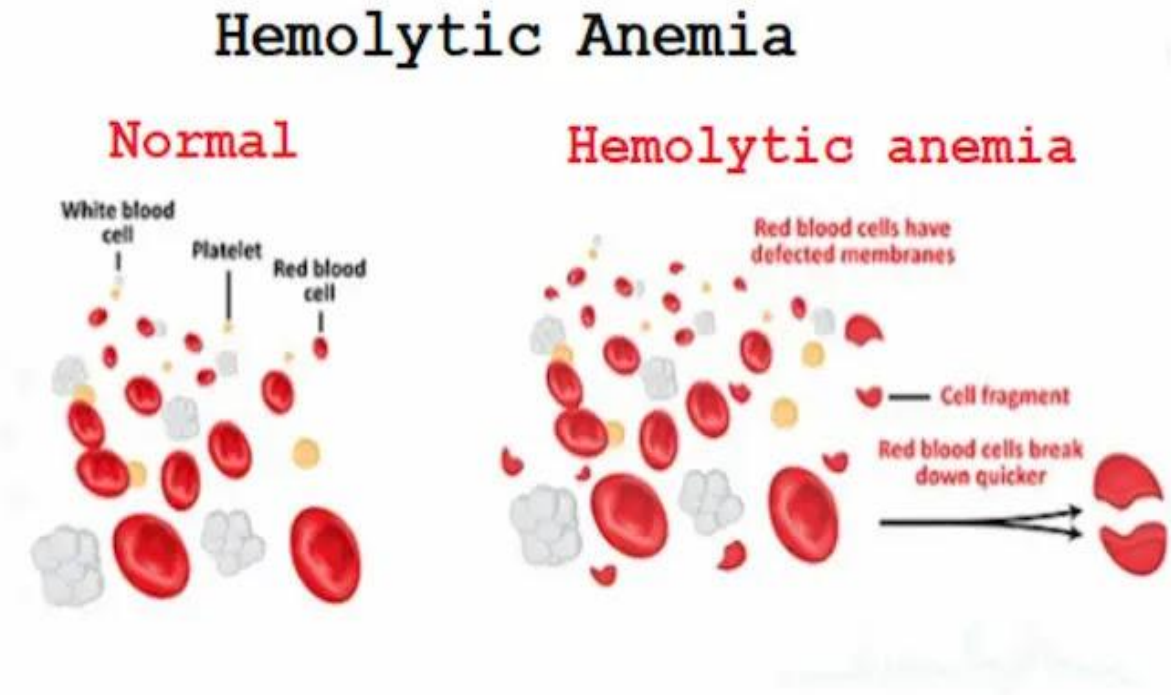
Hemolytic anemia can be due to many different causes: intrinsic and extrinsic mechanisms.

Lysis of erythrocytes (RBC) leads to spillage of their contents (LDH) into serum.

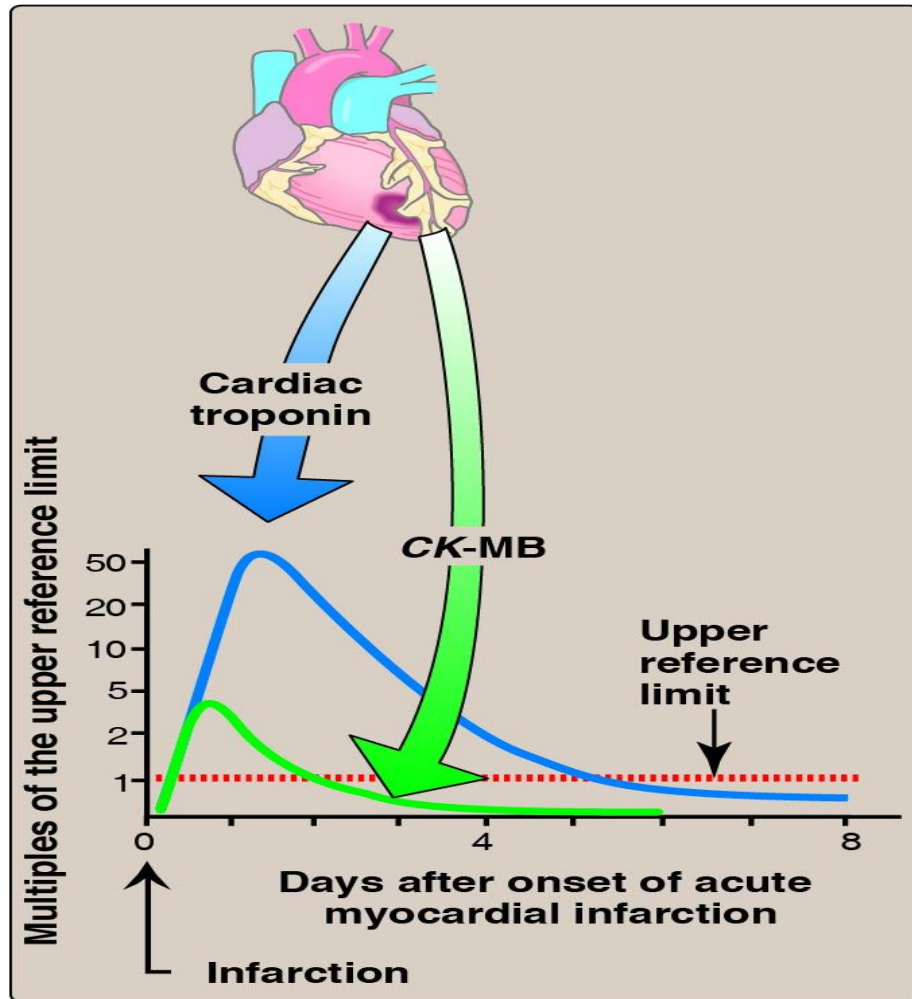
Hemolysis characterized by **increase in serum LDH**

Additional lab findings:

- Low hemoglobin (anemia)
- Increased indirect bilirubin (Jaundice)
- Reticulocytes (Precursors of RBC)



Diagnosis of Myocardial Infarction



Chest pain: Clinical sign (however not all patients have chest pain).

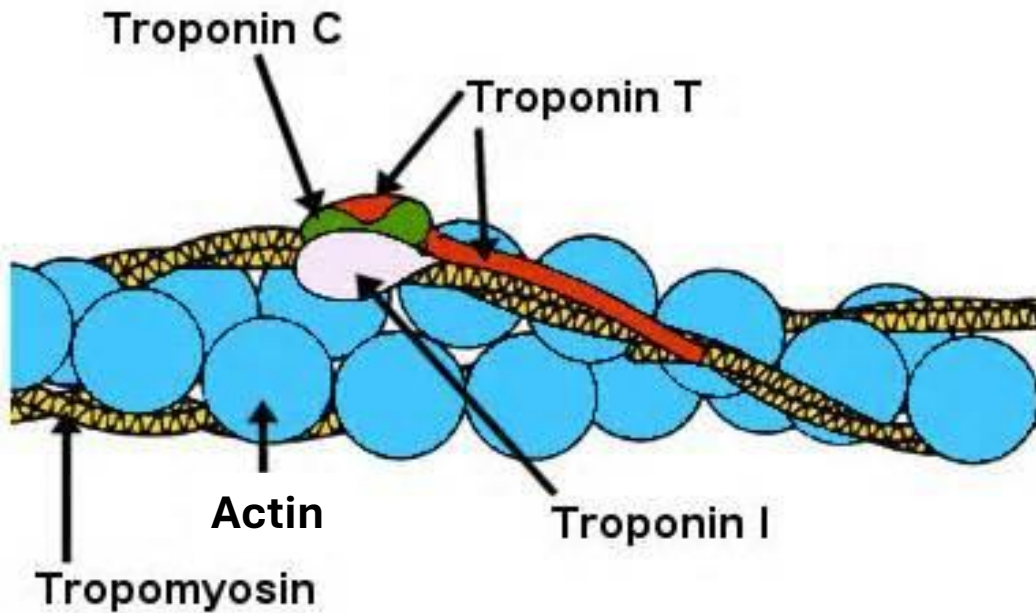
Electrocardiogram changes: Used to diagnose MI (STEMI)

Clinical presentation and ECG changes are considered **gold standard**

Cardiac markers:

- 1) Increase in cardiac troponins (cTnI or cTnT)
- 2) Increase in CK-MB
- 3) Myoglobin (early but not specific for heart)

Troponins



- **Troponins** are proteins involved in muscle contraction
- **Cardiac-specific isoforms** of troponins released into plasma in cardiac muscle damage
- **Serial measurements**
- **Reinfarction**

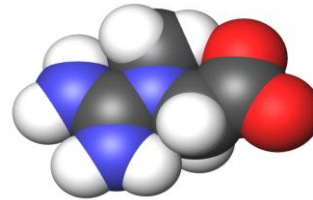
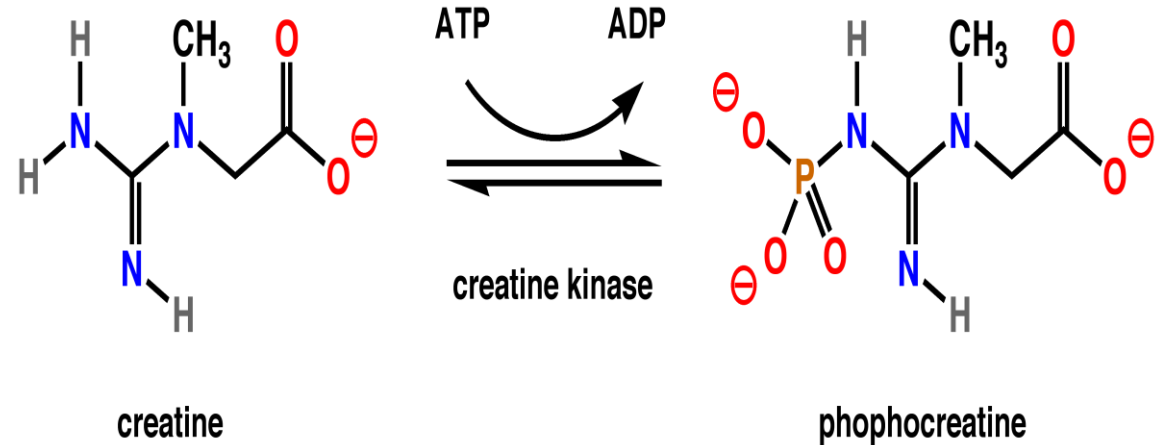
Three types:

- **Cardiac troponin I (cTnI: Inhibition of actomyosin ATPase)**
- **Cardiac troponin T (cTnT: Tropomyosin binding)**
- Troponin C (calcium binding) does not have specific isoform for the heart.

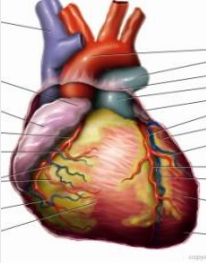
Creatine Kinase (CK)

■ Catalyzes a reversible reaction

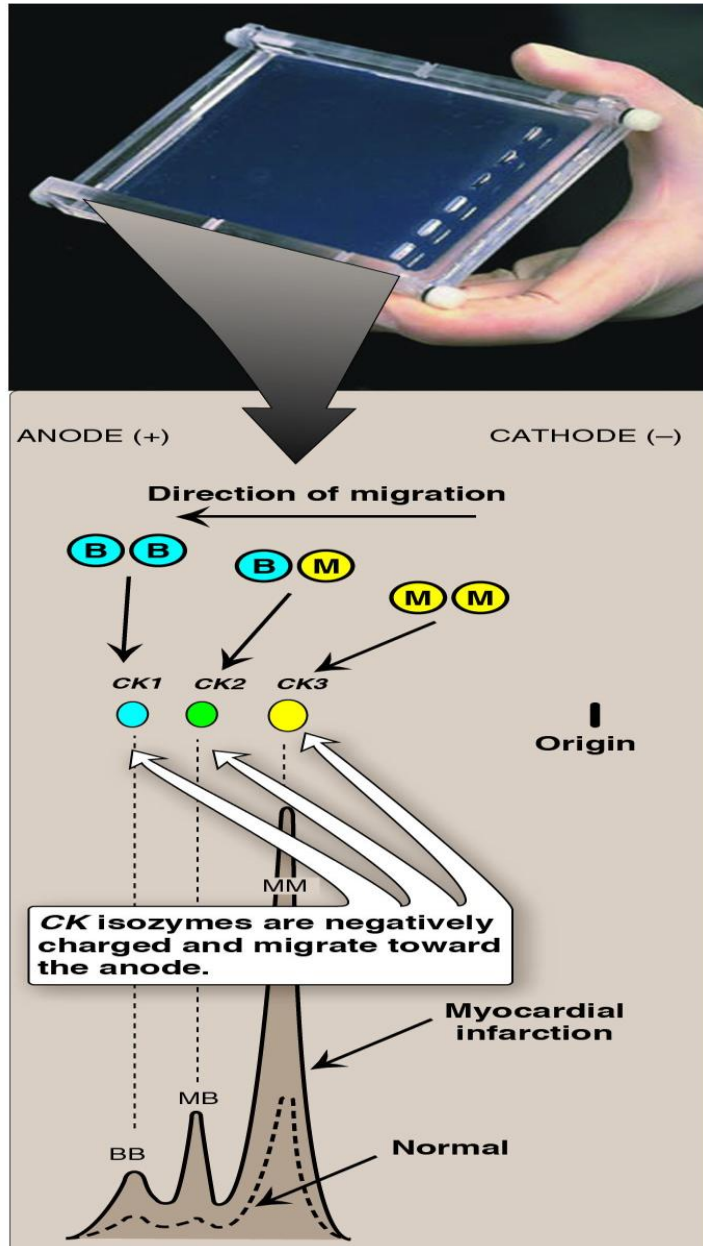
- At high **ATP** levels **creatine phosphate (CrP)** is formed.
- At high **CrP** levels **ATP** is formed using the energy-rich **CrP**
- Stores energy, which is
- Used by cells and tissues requiring high ATP.



Isoenzymes

ISOENZYME	COMPOSITION Contains 2 subunits (B or M)	TISSUE	CLINICAL CORRELATION
CREATINE KINASE 1 (CK-1)	CK-BB 	BRAIN	Serum CK-BB is elevated after CNS damage
CREATINE KINASE 1 (CK-2)	CK-MB 	CARDIAC MUSCLE TISSUE (highest ratio of CK-MB/CK-MM)	Serum CK-MB and total CK are elevated after myocardial infarction .
CREATINE KINASE 1 (CK-3)	CK-MM 	SKELETAL MUSCLE TISSUE (low ratio of CK- MB/CK-MM)	Serum CK-MM and total CK is elevated in rhabdomyolysis or muscular dystrophy .

Measurement of CK isoenzymes in the diagnosis of MI



Criteria for indication of an MI:

- **Elevated Troponins I or T**

A high CK-MB/total CK ratio indicates a large area of damaged heart muscle.

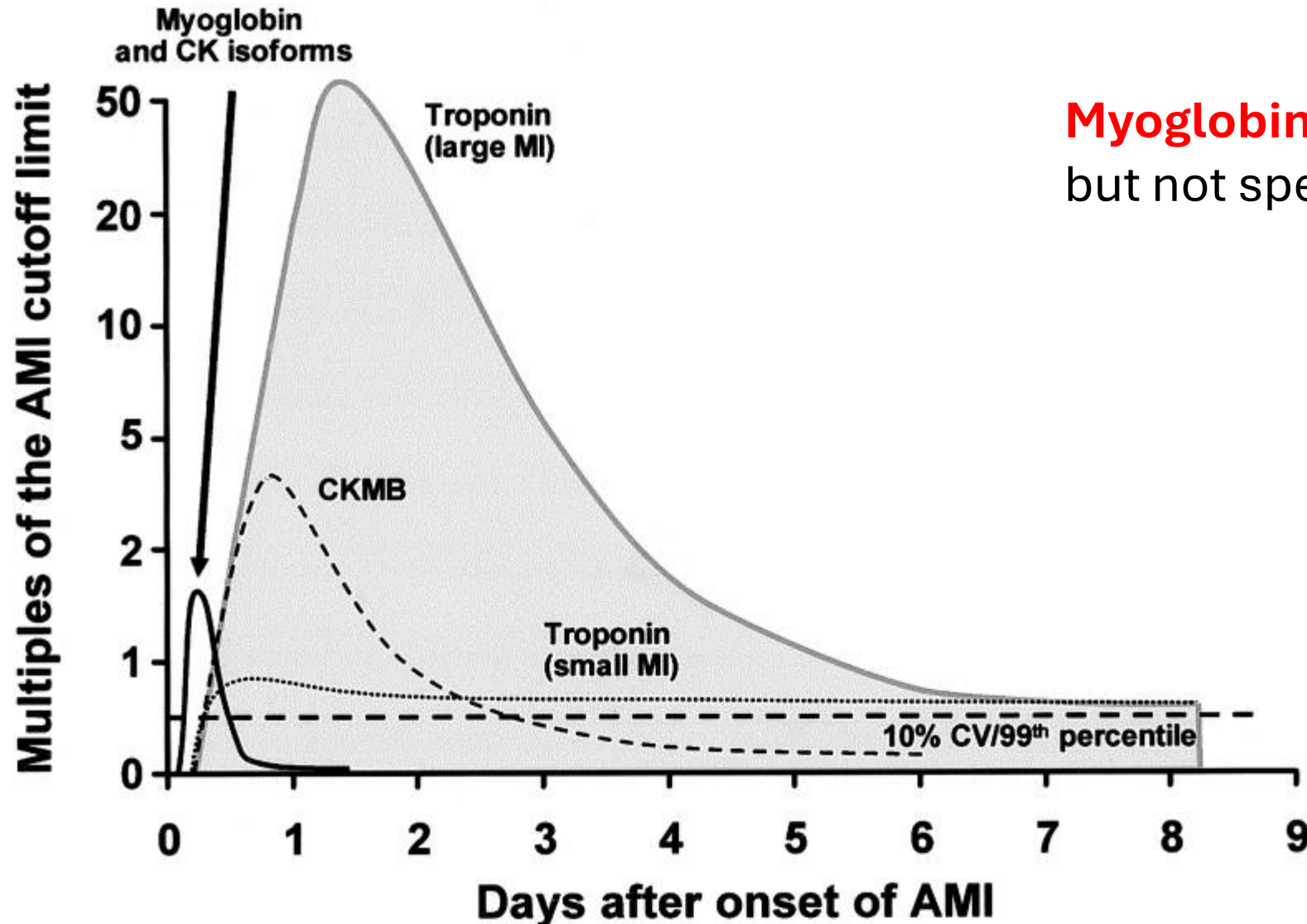
- The **CK-MB/total CK ratio** is $> 6\%$
- The **CK-MB/total CK ratio** is $> 3\%$ and troponins are increased.

- **Myoglobin** (nonspecific)

Measurement

- **ELISA (Radioimmunoassay):** are used to measure CK-MB, cTnI, cTnT and myoglobin.
- **Subunit composition, electrophoretic mobility** and enzyme activity of creatine kinase isoenzymes.

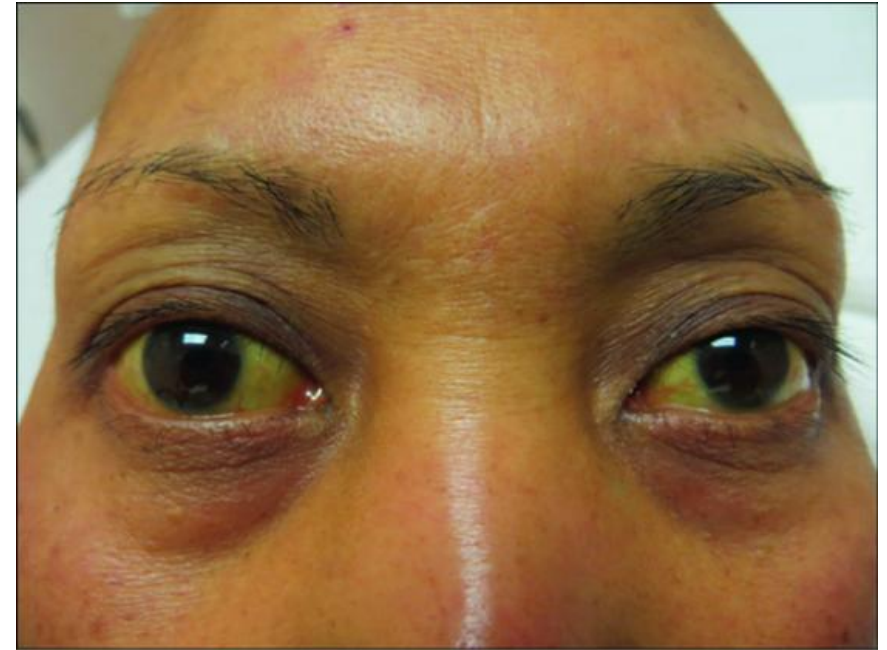
Time frame of MI injury markers CK-MB and troponin



Myoglobin- detected early (4 hours)
but not specific

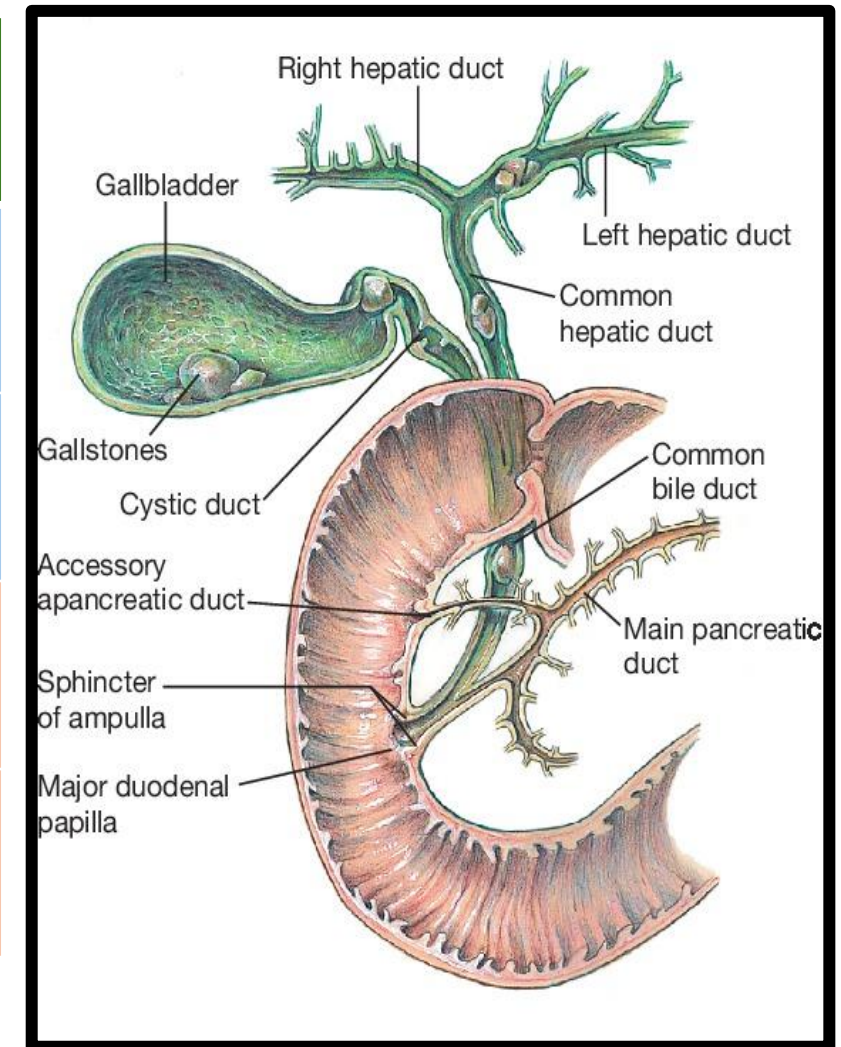
Liver disease: Markers

Liver function	Effects in liver pathology
Albumin	Decreased (biosynthetic marker)
Bilirubin	Increased (Jaundice)
Prothrombin time	Increased (leads to increased bleeding tendency)
Platelets	Decreased (Leads to increased bleeding tendency)



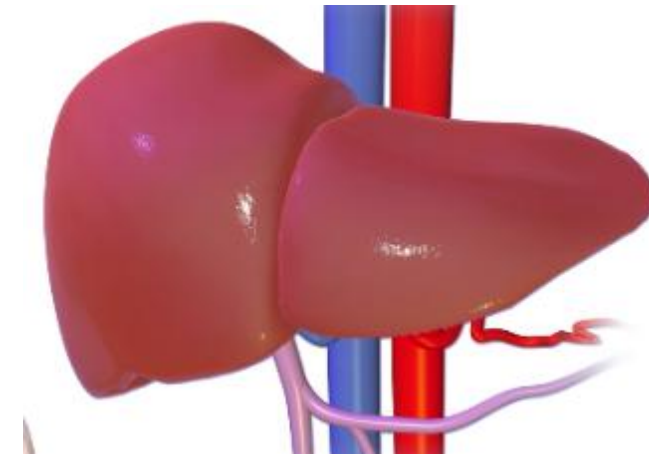
Liver and biliary pathology: Enzyme markers

Liver enzymes	Liver and biliary pathology
Aspartate aminotransferase (AST)	Increased
Alanine aminotransferase (ALT)	Increased
Alkaline phosphatase (ALP)	Increased (in cholestasis, bone disease)
γ - glutamyl transpeptidase (GGT)	Increased (liver and biliary diseases)



ALT/AST ratio: Indicator of type of liver damage

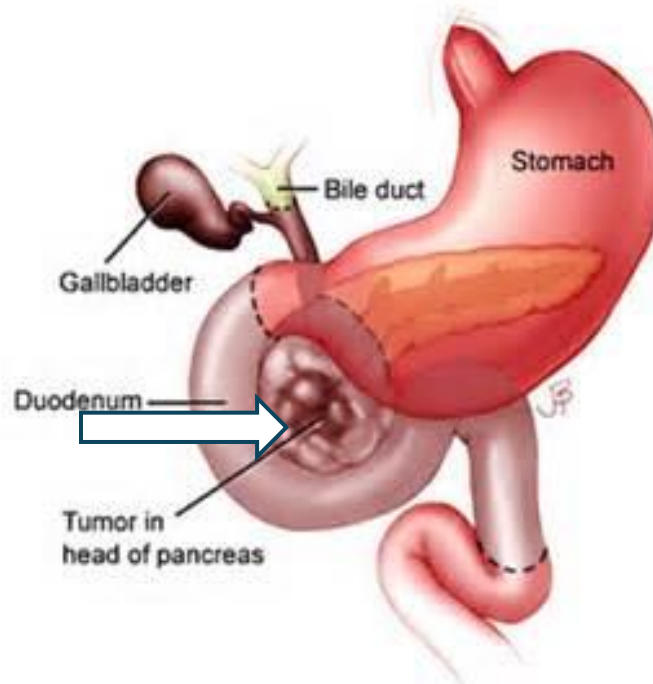
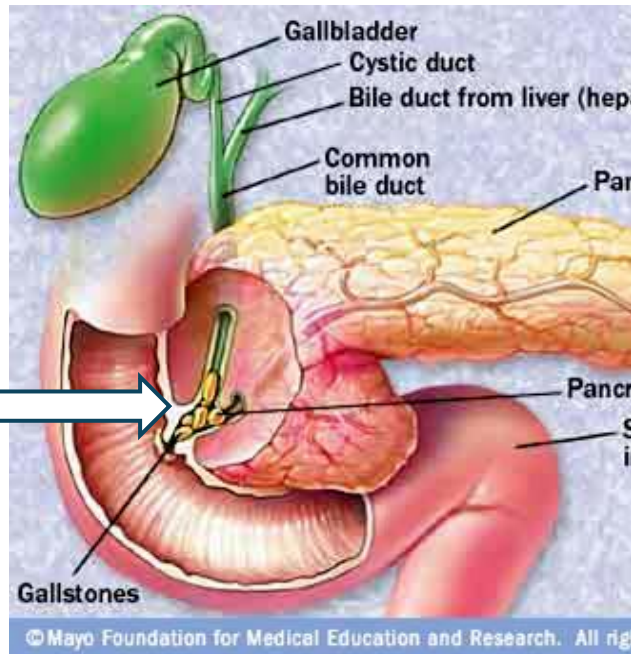
- **Acute liver disease**
 - **ALT>>>AST**: **Acute hepatitis (Acute liver disease)**
 - **ALT >>> AST (>1000 U/L)**: Drug induced liver injury, acute viral hepatitis, autoimmune hepatitis
- **Chronic liver disease**
 - **Aspartate aminotransferase (AST): Alanine aminotransferase (ALT) ratio >2:1 in Chronic alcoholic cirrhosis**
 - **AST>ALT** in **chronic** non-alcoholic liver disease



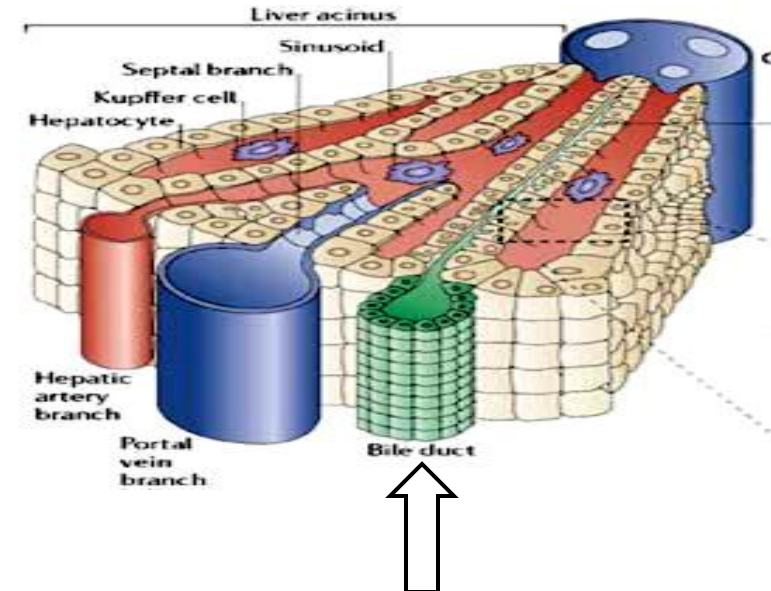
Bile duct obstruction

Extra-hepatic bile ducts obstruction:

- **Gallstones** (cholesterol gallstones)
- **Tumors** (tumor in head of pancreas)



Intra-hepatic bile ducts can be inflamed in liver cirrhosis.



Obstructive jaundice (biliary disease) markers

- **Alkaline phosphatase (ALP)**

- Found in high concentration in cells of the bile canaliculi and biliary ducts
- ALP cleaves phosphates from nucleotides and proteins which leads to the alkaline pH of bile.

- **γ -glutamyl transferase (GGT)**

- Bound to plasma membrane of hepatocytes
- GGT increase in serum indicates **injury** of **bile ducts** and **liver**.
- Elevated in gallstones, ethanol consumption, or medical drugs which induce liver enzyme synthesis

Pancreatitis

- **Acute pancreatitis**

- Autodigestion of the pancreas by pancreatic enzymes
- Due to premature activation of **pancreatic zymogens**

Causes:

- **Ethanol**
- **Gallstones**
- Idiopathic
- Trauma
- Steroids



- **Chronic pancreatitis**- chronic inflammation of the pancreas

Causes:

- Alcohol use disorder
- Genetic predisposition: Cystic fibrosis, hyperlipidemia.
- Idiopathic (cause not known)

Pancreatitis diagnosis

Acute pancreatitis:

- Acute epigastric pain radiating to the back
- Serum amylase or lipase (more specific) elevated three times above normal limit.
- Imaging findings (ultrasound, CT, MRI)

Ethanol induced: Serum lipase/amylase ratio greater than 2

Gallstone blockage: Elevated serum amylase, lipase, ALP and GGT

Chronic pancreatitis:

- Amylase and lipase may or may not be elevated
- Serum enzyme levels may be lower due to less synthesis by diseased pancreas

Summary:

Alcohol related damage of liver or pancreas

Chronic alcoholic cirrhosis:

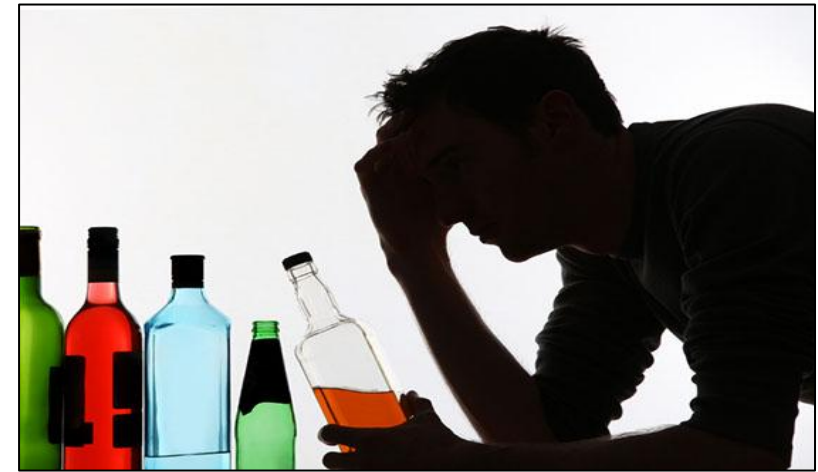
Serum **AST/ALT** ratio $>2:1$

GGT and **ALP** (Chronic cirrhosis)

Alcohol-induced Pancreatitis:

Serum **lipase** /**amylase** ratio >2

Elevated GGT



Alkaline phosphatase

is found in bone, bile ducts, placenta and intestine

- **ALP in bones** creates an **alkaline pH** for **calcium deposition** onto **collagen**.
- **Physiological increase in ALP** occurs in serum during bone growth (children) and during pregnancy.
- **Abnormal increase** in serum in **Paget disease, vitamin D deficiency (rickets)** or **bone tumors**.
- **Note:** High serum ALP is also found in bile duct obstruction, but GGT levels are also elevated, and signs of liver disease are present

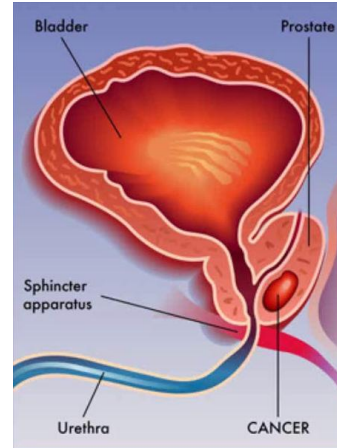


Paget's disease



Tumor markers

Prostate malignancy



Prostate antigen test (PSA)

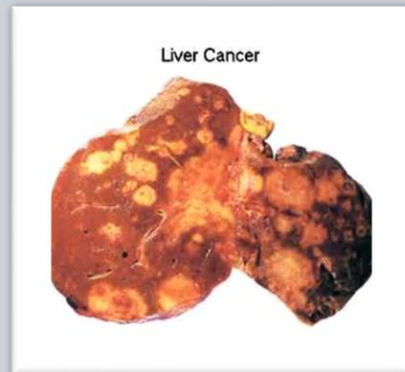
Acid phosphatase

Prostate inflammation:

- Prostate malignancy
- Benign prostate hyperplasia
- Prostatitis



Liver malignancy



Alpha-fetoprotein (AFP)

Normally synthesized in fetal liver (instead of albumin).

Pregnancy

Elevated AFP in adults can indicate **liver cancer**.

Summary of injury markers in serum

Disease	Injury marker in serum	Special
Heart disease including Myocardial infarction	Cardiac Troponins I and T (TnI and TnT)	Used as early injury markers and are also used up to 7-10 days after acute injury.
	Total Creatine kinase (CK) and CK-MB	CK-MB/total CK ratio indicates severity of damage. Enzymes are released into cytosol after cell death.
	Time frame after MI: Early measured injury markers are myoglobin together with CK-MB and troponins .	Myoglobin is non-specific but supports diagnosis when CK-MB and troponins are elevated.
Muscular diseases rhabdomyolysis, dystrophy	Total CK and CK-MM	
Liver disease: Viral infection Alcohol damage Liver cirrhosis	Aspartate aminotransferase (AST) Alanine aminotransferase (ALT) Alkaline phosphatase (ALP) from intrahepatic bile ducts and Gamma-glutamyl transpeptidase (GGT) from liver membrane often in response to alcohol	Acute liver disease: ALT>AST levels Chronic alcoholic cirrhosis: AST/ALT ratio is larger than 2, GGT Cirrhosis: ALT, AST, GGT and ALP
Biliary disease	Alkaline phosphatase (ALP) Gamma-glutamyl transpeptidase (GGT) (both are increased)	ALP and GGT are released from bile ducts and from inflamed cell membranes of hepatocytes.
Pancreatic disease (gallstones or excessive alcohol)	Alpha-amylase and pancreatic lipase Due to gallstones: in addition elevated ALP and GGT Due to alcohol: in addition Lipase/amylase ratio larger than 2	Acute pancreatitis leads to elevated levels in the blood. Chronic pancreatitis can lead to lower levels than acute pancreatitis.
Bone disease (Paget disease, rickets, bone tumor)	Alkaline phosphatase (ALP) (note: only ALP increased, but not GGT)	High ALP is normal in growing bones or healing or during pregnancy
Cancer of the prostate	Prostate specific antigen test (PSA)	PSA substituted the test using acid phosphatase
Cancer of the liver	Alpha-fetoprotein for liver cancer	Alpha-fetoprotein is found instead of albumin in the fetus

**Thank you, if you have questions please
email me.**

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